MaleVis Related-Work Table and Research Gap Analysis

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Related Work: Malevis with CNN and Attention

The MaleVis dataset represents malware and benign software classes as three-channel RGB color images so that recurring byte-level visual patterns can be learned directly by image-classification models. The studies below were selected because they use convolutional neural networks, channel or spatial attention, or hybrid attention mechanisms tested on MaleVis or similar multi-class image configurations. This review focuses on the model designs, attention mechanics, evaluation protocols, reported accuracies, and limitations that are directly relevant to a Custom CNN + Attention framework, with a major focus on handling the redundant byte-image problem.

1. Yao et al. (2025) **[1]** proposed a Res2NeXt-based malware-classification framework incorporating a lightweight Adaptive Channel Attention (ACA) mechanism. The attention module emphasizes informative feature channels while reducing the influence of irrelevant information in malware RGB images. The authors evaluated the approach on MMCC and MaleVis, reporting accuracies of 99.26% and 98.04%, respectively. The study also used confusion matrices, t-SNE, and Grad-CAM visualizations to examine classification performance and learned image features. However, the specialized Res2NeXt architecture leaves room to investigate whether a simpler custom CNN with lightweight attention can achieve competitive results with fewer parameters and lower inference latency. A controlled comparison between the same CNN with and without attention would also help quantify the contribution of the attention mechanism.

2. Fan et al. (2025) **[2]** presented GCSA-ResNet, a malware-classification architecture that integrates a Global Channel-Spatial Attention module into ResNet-50. The module combines channel attention, channel shuffling, and spatial attention to emphasize informative features and improve feature interaction. The model reported accuracies of 99.08% on Malimg and 98.58% on Microsoft BIG 2015, demonstrating the potential of combining channel and spatial attention for malware classification. However, the architecture relies on a comparatively complex ResNet-50 backbone, and its results do not establish whether a smaller custom CNN can achieve a similar balance between classification performance and computational efficiency. Additionally, evaluation on MaleVis would help assess its applicability to the dataset assigned in the present project.

3. Ben Abdel Ouahab et al. (2023) **[3]** investigated Vision Transformers (ViTs) for malware classification and compared them with conventional CNN models on the Malimg dataset. The ViT divides each malware image into patches and employs multi-head self-attention to capture long-range dependencies between different image regions. This approach enables the model to learn global relationships that may not be captured effectively by local convolutional operations alone. However, Vision Transformers can require greater computational resources and training time than conventional CNNs. Therefore, the present project can investigate whether a lightweight attention mechanism integrated into a custom CNN can capture informative malware-image patterns while maintaining lower computational overhead.

4. Kim et al. (2023) **[4]** advanced an attention-based cross-modal CNN that combines two representations derived from malware binaries: malware images and structural-entropy sequences. Cross-modal attention allows each modality to reinforce the other before higher-level feature extraction. Using 10-fold cross-validation, the model achieved 99.09% accuracy and an F1-score of 0.976 on a filtered Malimg dataset containing 8,250 samples. However, the approach requires an additional structural-entropy representation, making the model more complex than an image-only classifier. Furthermore, the contribution of attention is mixed with the benefits of incorporating a second modality, leaving room to investigate attention-based classification using malware images alone.

5. Van Dao et al. (2022) **[5]** outlined a lightweight feature-synthesis pipeline that combines a small CNN with an attention-enhanced variational autoencoder (VAE). The attention component follows CBAM principles by using channel and spatial attention to refine the learned representations. The extracted features were classified using several conventional classifiers, with Random Forest achieving 99.40% accuracy on the imbalanced Malimg dataset under 10-fold cross-validation, while the balanced setting achieved 98.40%. A key limitation is that the improvement cannot be attributed to attention alone because the pipeline combines CNN features, an attention-enhanced VAE representation, and an external classifier rather than comparing the same end-to-end CNN with and without attention.

Summary Table

Paper	Year	Dataset / Protocol	Method	Attention Type	Main Result
[1] Yao et al.	2025	MaleVis and MMCC	Res2NeXt with Adaptive Channel Attention (ACA)	Lightweight adaptive channel attention	98.04% accuracy on MaleVis; 99.26% on MMCC
[2] Fan et al.	2025	Malimg and Microsoft BIG 2015	GCSA-ResNet based on ResNet-50	Global channel-spatial attention	99.08% accuracy on Malimg; 98.58% on Microsoft BIG 2015
[3] Ben Abdel Ouahab et al.	2023	Malimg; ViT compared with conventional CNN	Vision Transformer (ViT) and CNN baseline	Multi-head self-attention	ViT: 98.38% accuracy; CNN: 97.90%
[4] Kim et al.	2023	Filtered Malimg (8,250 samples); 10-fold cross-validation	Cross-modal CNN using malware images and structural-entropy sequences	Cross-modal attention	99.09% accuracy; F1-score = 0.976
[5] Van Dao et al.	2022	Malimg; 10-fold cross-validation; imbalanced and reduced balanced settings	Lightweight CNN + attention-enhanced VAE features + Random Forest	CBAM-style channel and spatial attention	99.40% accuracy on imbalanced Malimg; 98.40% on reduced balanced data

Research Gap

Recent literature demonstrates that convolutional neural networks and attention-driven architectures provide powerful representations for image-based software classification. However, a major limitation in existing frameworks is that attention modules are frequently integrated into massive, over-parameterized backbones (such as ResNet-50 or heavy Vision Transformers). This architectural complexity masks the standalone contribution of the attention mechanism, making it difficult to isolate whether performance gains stem from the attention block or simply from increased model capacity. Furthermore, a reliance on raw accuracy as the primary evaluation metric fails to reflect whether these models are computationally efficient or capable of ignoring uninformative region properties, such as trailing canvas padding bytes.

To address this gap, this study deploys a streamlined, custom CNN backbone integrated with a lightweight channel-and-spatial attention module specifically optimized for the MaleVis dataset. This design isolates and explicitly evaluates the attention module's ability to filter out redundant byte-image regions. To establish clear empirical verification, the proposed model will be compared directly against an identical custom CNN control baseline operating without attention under uniform training conditions. Both models will be rigorously benchmarked using overall accuracy, macro-F1 score, class-wise recall metrics, total parameter footprints, and localized inference latency profiles.

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